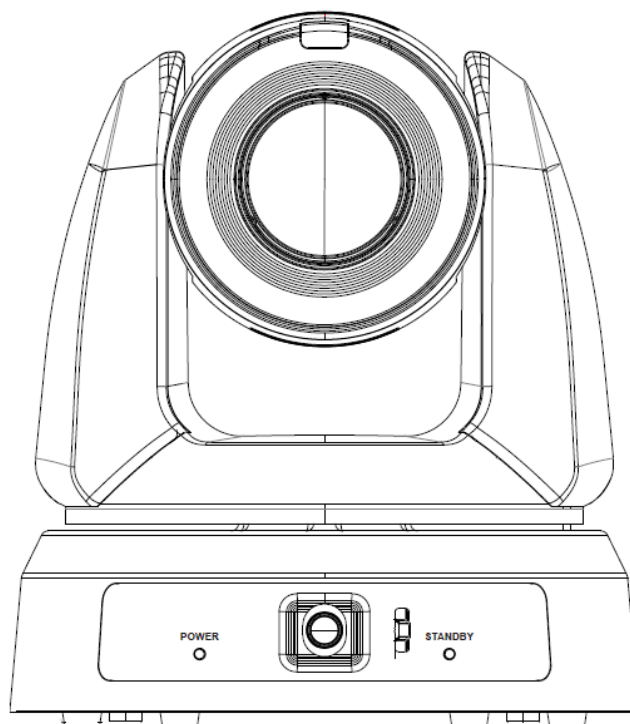


VC-TR61/ VC-TR61N

Auto Tracking Camera

User Manual - English



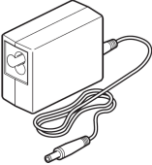
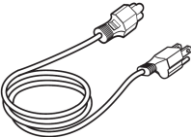
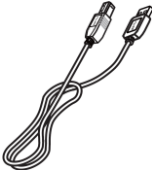
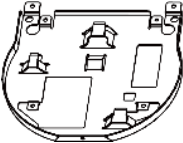
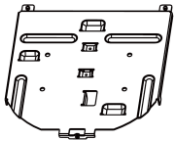
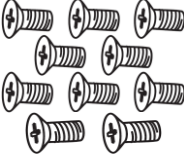
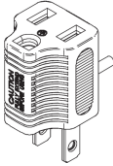
To download the latest version of Quick Start Guide, multilingual user manual, software, or driver please visit the Lumens website:

<https://www.MyLumens.com/support>

Contents

Chapter 1 Packing List.....	2
Chapter 2 Product Appearance	3
2.1 I/O interface	3
2.2 LED Indicator.....	4
2.3 Tally Lamp	4
Chapter 3 Installation Instructions	5
3.1 Scenario.....	5
3.2 Instruction for installation.....	5
3.3 Connection	8
Chapter 4 Remote Control and Setting Menu	9
4.1 Remote control	9
4.2 Setting Menu	10
Chapter 5 Network setting and Web interface	14
5.1 Connecting Camera to Network.....	14
5.2 Web interface functions	17
Chapter 6 Troubleshooting	33
Chapter 7 Addendum.....	34
Chapter 8 Safety Instructions.....	37
Copyright Information	38

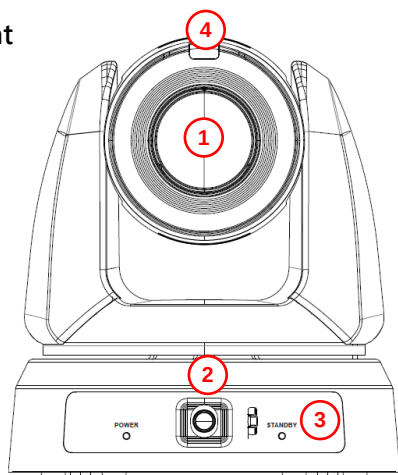
Chapter 1 Packing List

Power Adapter	Power Cord	USB 3.0 Cable-2m (Type B - Type A)	Remote Control
	 Appearance may vary depending on country/ region		
Metal Plate A	Metal Plate B	M3 Screws Silver x8/ Black x2	3 PIN-2 PIN Connector
			 (For Japan only)

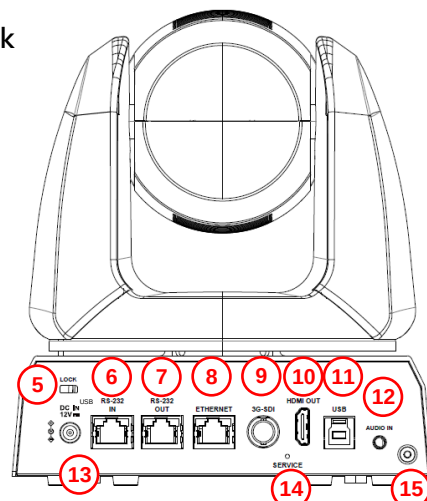
Chapter 2 Product Appearance

2.1 I/O interface

Front



Back



NO.	項目	功能說明
1.	Main camera lens	PTZ camera lens
2.	Panoramic lens	Panoramic camera lens Note When the camera is installed at a height above 2 meters , we recommend adjusting the tilt position. Refer to <u>3.2.5 Tilting the Panoramic Lens</u>
3.	LED indicators	Displaying the status of the camera, refer to <u>2.2 LED Indicator</u>
4.	Tally Lamp	Displays the Tally Lamp status of the camera
5.	Lock	Safety lock slot
6.	RS232 IN	RS-232 port, up to 7 cameras can be connected.
7.	RS232 OUT	Pin out please refer to <u>3.3.2 RS-232 Pin Definition</u>
8.	Ethernet	Network port, with PoE+ (IEEE802.3at) support
9.	3G-SDI	3G-SDI output (BNC 75Ω / supports SMPTE 425M)
10.	HDMI OUT	Video output
11.	USB3.0	Connect to the computer's USB port; delivers USB video
12.	Audio In	3.5mm Line in
13.	DC IN 12V	DC supply connecting port
14.	Service	Safety lock slot
15.	Lock	Safety lock slot

2.2 LED Indicators

Status	Power	Standby
Power not connected	No light	No light
Startup in progress (initialization)	Green	Orange
Power on (In use)	Green	No light
Standby	No light	Orange
Tracking / Framing mode	Green Flickering	Green Flickering
Privacy mode	Orange	No light

2.3 Tally Lamp

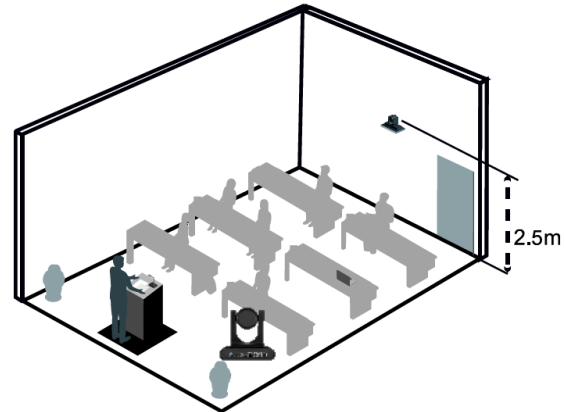
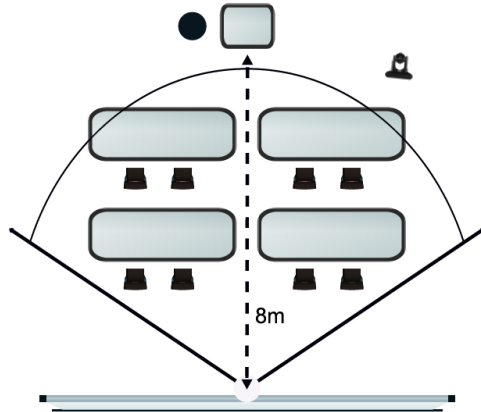
The Tally lamp function can be enabled by an RS-232 command. Please refer to the RS-232 command set document

Chapter 3 Installation Instructions

3.1 Scenario

VC-TR61/ VC-TR61N supports a tracking range of 2-18m

Mounting height: 2 - 3 m, best height recommended: 2.4 m



Note *For best tracking results, avoid placing images of people on posters or screens behind the subject.

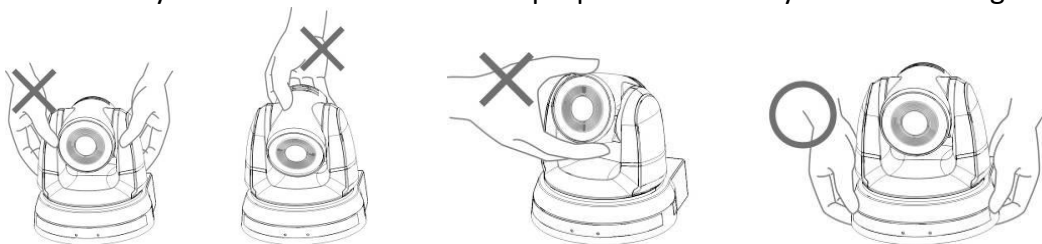
3.2 Installation Instructions

PTZ installation and connection requires special skills. It is important to securely install the device. Pay attention to your safety and to the safety of those around you during the installation. Ensure the safety of the installation environment. Do not install the device to an unstable ceiling or at a height that may risk injury to people below.

- 3.2.1 Please ensure installation safety, please do not mount the device on an unstable ceiling or in any location where it may fall, as this could lead to accidents.
- 3.2.2 Please check that accessories in the box are complete. Please contact the supplier for any shortage, and make sure to keep the accessories in the box.
- 3.2.3 Please choose an appropriate location for installation of camera in advance. Please determine an installation site according to the following requirements:
 - (A) Confirm the position for the object to be captured.
 - (B) Confirm whether the camera is set at a proper distance from light sources

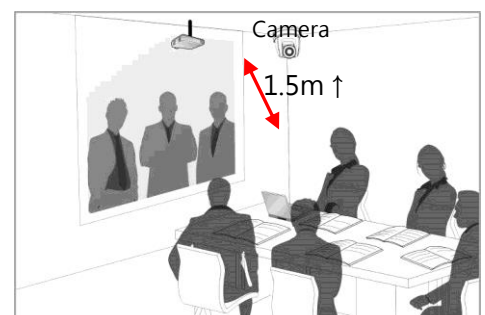
3.2.4 Handling Instructions

- Do not hold the camera by its pan/tilt head.
- Do not manually rotate the camera head. Improper rotation may result in damage.



- Before installation, confirm the camera orientation relative to the object to be captured.

- It is recommended that the camera should be set at a distance of more than 1.5 meters from the object to be captured. Adjust the position to achieve your required shooting angle.

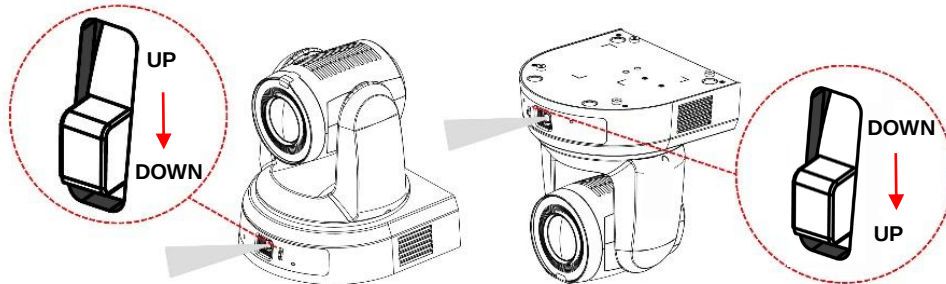


- The Camera (including metal plates) weighs approximately 2.7 kg. For ceiling mount, use a hanger that has obtained UL security approval to prevent the unit from falling down.
- Regularly check the stability and strength of the camera installation.

3.2.5 Tilting the Panoramic Lens

The panoramic lens supports three tilt positions: **Up / Normal / Down**

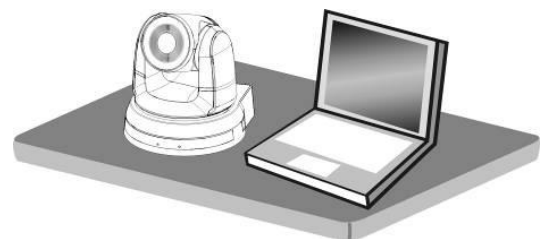
- When the camera is installed at a height above **2 meters**, it is recommended to adjust the angle of the lens.
- Upright Installation: Adjust to the **[Down]** position.
- Inverted Installation: Adjust to the **[Up]** position.
- After adjusting the tilt angle, please go to **[Tracking]> [Lens Tilt Direction]** and select the corresponding value in the web interface.



*For inverted installation, please enable the mirror/ flip function.

3.2.6 Installing the camera on the desk

Place the camera on a flat desk to ensure normal operation.



3.2.7 Mounting the camera to the ceiling

(A) Prepare the following items:

- Accessories in the box (metal plates A, B, M3 screw silver x 8, black x 2)
- 4x screws for locking the ceiling mounted hanger
- Drilling, screwdriver and ladder
- For dimensions of the camera and metal plate, please refer to [Chapter 7 Addendum](#)

(B) Installation steps

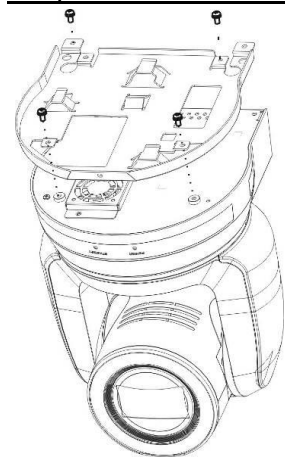
1. Fix the metal plate A to the camera base with
2. 4x M3 silver screws

3. Lock the metal plate B on ceiling mounted hanger

※Caution :

- Please use a hanger that has obtained UL security approval.
- Please ensure space for cable access.

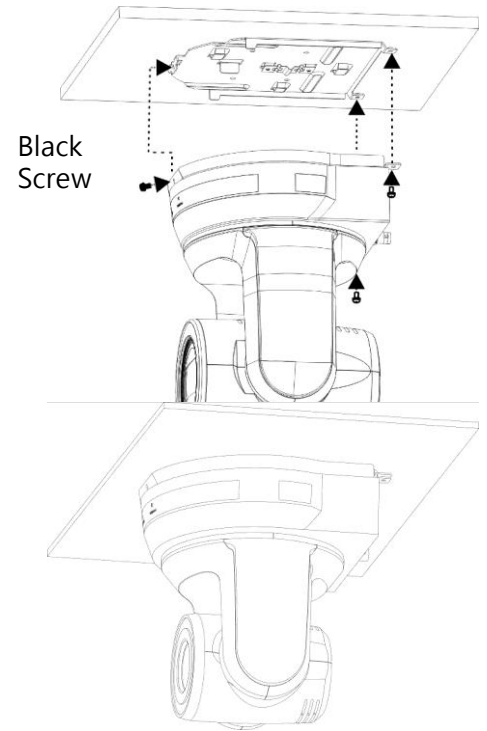
4. Combine metal plate A and metal plate B



- (1) Push metal plate A up to the ceiling and then to the right to latch onto metal plate B
- (2) Then secure with 2x M3 silver screws and 1x M3 black screw

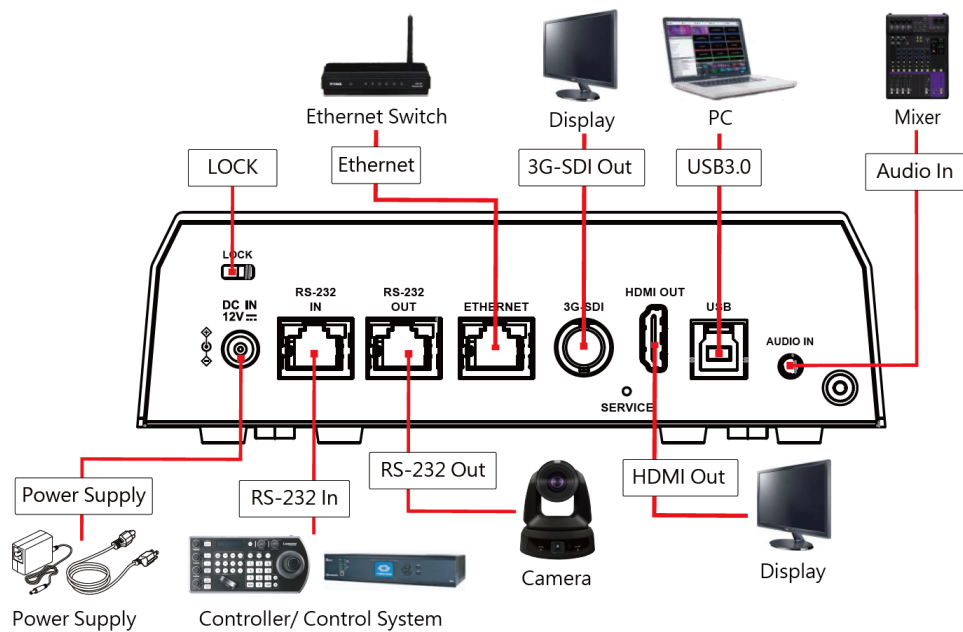
3.2.1 How to remove

1. Remove the connecting cables from the camera
2. Loosen the three screws that fix the metal plates A and B and push to the left to remove the camera
3. Remove the three screws



3.3 Connection

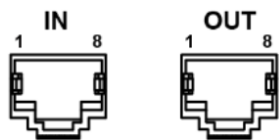
3.3.1 Connecting devices



Note Ethernet port supports PoE+ (IEEE802.3at)

3.3.2 RS-232 Pin Definition

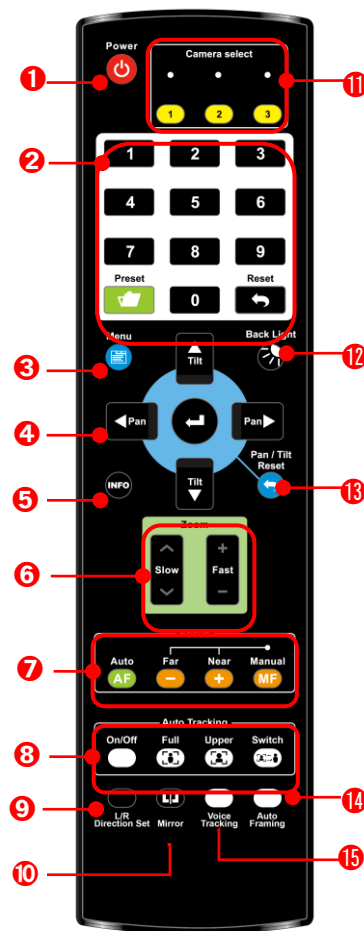
Note Daisy Chain for up to 7 connected cameras



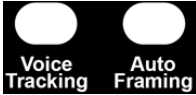
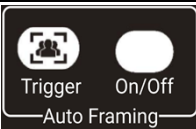
Pin No.	IN	OUT
1	NA	NA
2	NA	NA
3	GND	GND
4	TX	RX
5	RX	TX
6	DTR	DSR
7	NA	NA
8	NA	NA

Chapter 4 Remote Control and Setting Menu

4.1 Remote control



No.	Item	Description
1.	Power	Power on / off
2.	Preset	0-9: Click to call the preset - Preset: Assign a preset number (0-9) to save the current position - Reset: Select a preset number (0-9) to clear its position
3.	Menu	Display OSD menu
4.	Pan/ Tilt/ Enter	- Pan / Tilt: Move the camera head - Home / Enter: Go back to Home position / Execute
5.	Info	Status information
6.	Zoom	Fast: high speed zoom / Slow: low speed zoom
7.	Focus	- Auto: Auto focus - Manual: Enable Manual Focus - Far/ Near: Adjust the focus in Manual mode
8.	Auto Tracking	- On/Off: Enable/Disable Auto Tracking - Full: Frames presenter's full body - Upper: Frames presenter's torso - Switch: Switch target to a different presenter
9.	L/R Direction Set	L/R Direction / Normal
10.	Mirror	Rotate the image (OFF / Mirror / Flip / Rotate)
11.	Camera select	Choose Camera ID 1 - 3
12.	Back Light	Turn on/off back light compensation
13.	Pan/Tilt Reset	Clear the Pan/Tilt setting
14.	Auto Framing (or ON/ OFF)	Enable/ Disable Auto framing ※Please refer to Note 1 for details.
15.	Voice Tracking (or Trigger)	VC-TR61/ VC-TR61N not supported ※Please refer to Note 1 for details.

New version

Previous version


※**Note 1:** The labeling of buttons #14/ #15 may vary depending on the remote control version.

Please refer to the image on the left for details.
(The labeling differs, but the functions are the same.)

#14 Auto Framing = Auto Framing On/Off

#15 [Voice Tracking] or [Trigger]:

- When used with a camera that supports voice tracking: on/ off the voice tracking function.
- When used with a camera that does not support voice tracking:

- Single-lens: Auto framing will reset to wide angle and reframe automatically.
- Dual-lens: No corresponding action.

※Dual-lens cameras feature a built-in wide-angle lens for improved framing, eliminating the need for manual calibration

4.1.5 Preset numbers to avoid

Some preset numbers are reserved for specific functions. Avoid using these numbers for regular position presets.

Preset Number	Description
80/81	Turns tracking or framing on or off, depending on the selected mode.
109	Return the PTZ lens to a centered wide view, or to the user-defined Preset 109 position.
110	When tracking starts, the system automatically saves the current PTZ position as Preset 110 for the 'Back to Start Position' target-lost action.
111	The system stores the Stage Zone position as Preset 111.
101~104	Preset for Partition Tracking zone 1~4
105~108	Preset for Partition Framing zone 1~4
200	Custom Center Stage Position

4.2 Setting Menu

Press [MENU] on the remote control to call the on-screen menu. Default is indicated in bold.

Layer 1	Layer 2	Layer 3	Description
Exposure	Mode	Full Auto / Shutter Priority/ Iris Priority/ Manual/ White Board	Exposure mode setting
	Exposure Comp	On/ Off	Turn on/off Exposure Compensation
	Exposure Comp. Level	-6- C -4	Adjustable after Exposure Comp. is activated
	Spot Light	On/ Off	Turn on/ off Spot Light
	Spot Light Position	X(0-6)Y(0-4)	Adjustable after Spot Light is activated
	Gain Limit	8- 30 dB	Max. limit value of electron gain
	WDR	Off / 1/ 2/ 3	WDR settings
	Anti-Flicker	50Hz/ 60Hz/ Off	Activate the anti-flicker function to eliminate flicker from lighting.
	Shutter Speed	1/10000-1/25	Adjustable when the Exposure mode is set to Shutter Priority or Manual
	Iris Level	F1.8- C -F11/ Off	Adjustable when the Exposure mode is set to IRIS Priority or Manual
	Gain Level	0 ~45dB	Adjustable when the Exposure mode is set to Manual
White Balance	Mode	1.Auto 2.Indoor 3.Outdoor 4.One Push WB 5.ATW 6.Manual 7.Sodium Lamp	Select the color temperature 1. 4000k - 7000k 2. 3200k 3. 5800k 4. 1700k - 10000k 5. 1700k - 10000k 6. Manual 7. 2800k
	One Push Trigger	ENTER	One push trigger

Layer 1	Layer 2	Layer 3	Description
	Manual Red	0- <u>C</u> -128	Adjustable when the white balance mode is set to Manual
	Manual Blue	0- <u>C</u> -128	Adjustable when the white balance mode is set to Manual
Tracking	Type	Auto Tracking / Auto Framing	
Picture	3D NR	Off / Low/ Typ / Max	3D noise reduction
	Image Mode	Default / Custom	Users can customize the selected image mode
	Gamma	0- <u>C</u> -3	Adjustable when Image Mode set to Custom
	Brightness	0- <u>C</u> -25	Adjustable when Image Mode set to Custom
	Contrast	0- <u>C</u> -25	Adjustable when Image Mode set to Custom
	Saturation	0- <u>C</u> -25	Adjustable when Image Mode set to Custom
	Sharpness	0- <u>C</u> -11	
Pan Tilt Zoom	Pan/Tilt Limit	On/ Off	On/ Off Set angle limit
	Pan Right Limit	0- 170	Limit the panning angle (right side)
	Pan Left Limit	- 170 -0	Limit the panning angle (left side)
	Tilt UP Limit	0- 90	Limit the tilt angle (up)
	Tilt Down Limit	- 30 -0	Limit the tilt angle (down)
	Pan Flip	On/ Off	Reverses the Pan direction
	Tilt Flip	On/ Off	Reverses the Tilt direction
	Preset Speed	5/ 25/ 50/ 80/ 120 deg/sec	Set the robotic and zoom speed of the camera
	PTZ Speed Comp	On/ Off	Enable the zoom speed independent from the robotic speed
	D-Zoom Mode	On / Off	Set the digital zoom limit
	Limit	2x -8x	Note Digital zoom not supported with: ■ 4K resolution ■ 1080P/ 720P, Stream/ UVC outputs
D-Effect	Mirror	Off / Mirror/ Flip/ Mirror+Flip	Activate image rotation and/or mirroring
Auto Focus	AF Sensitivity	Low/ Middle / High	Changes the sensitivity of the auto-focus
	AF Frame	Center Area/ All Area	Changes the focusing zone to the selected mode ■ Central area: focusing will be performed based on the center ■ Full Frame: focusing will be calculated based on the full screen
	Zoom tracking	On/ Off	Focus mode during lens zooming (It can be set to Manual Focus (MF) mode)

Layer 1	Layer 2	Layer 3	Description
	PTZ Assist	On/ <u>Off</u>	Available when Focus is in Manual Mode Note PTZ Assist is triggered only when the camera has moved in any axis. It will not dynamically respond to changes in the scene.
Ethernet	DHCP	<u>On</u> / Off	Enable/Disable dynamic host configuration
	IP Address	<u>192.168.100.100</u>	Configurable when DHCP is set to Off
	Subnet mask	<u>255.255.255.0</u>	
	Gateway	<u>192.168.100.254</u>	
Audio	Audio In	<u>Line In</u>	3.5m Line In
	Audio Enable	On/ <u>Off</u>	Turn on/off audio output
	Audio Volume	0- <u>C</u> -10	Volume Setting
	Audio Delay	On/ <u>Off</u>	When the audio is not synchronized with the video, enable this function to set an audio delay time
	Audio Delay Time(ms)	<u>-10</u> ~-500ms	Set audio delay time
	Encode Sample Rate	<u>48</u> (AAC)	Set the encode type and sample rate
System	Prompt	On/ <u>Off</u>	Turn on/off the prompt information on the display
	IR Receive	<u>On</u> / Off	Turn On/off the infrared reception. When off, the remote control will not be able to control the camera. Re-start the camera to re-activate.
	IR Select	<u>1</u> / 2/ 3	Pair the remote controller with the camera.
	Tally Lamp	On/ <u>Off</u>	Enables the camera's tally lamp.
	Tally CMMD Mode	Normal/ <u>Link</u>	Set whether the tally lamp is linked with RS232 commands. ■ Normal (Not linked): The tally lamp and mode are set separately. ➤ When the tally lamp is on but the default mode is off, an additional command is required to set the indicator light mode ➤ When the tally lamp is off, setting the mode will not trigger the indicator light ■ Link: ➤ When the tally lamp is on, the default mode is a red light ➤ When the tally lamp is off, setting the mode will turn on the tally lamp.

Layer 1	Layer 2	Layer 3	Description
	Tracking Led Status	On/ <u>Off</u>	When enabled, LED on the front panel will flash green while the camera is tracking.
	Language	<u>English</u> / 繁中/ 简中	
	Initial Position	<u>Last MEM</u> / 1st Preset	After turning on the camera, you can select the lens to be returned to <u>Last MEM</u> or 1st Preset Note 1st Preset =Preset 0
	Motionless Preset	On/ <u>Off</u>	The image will freeze when Preset is executed. Freeze will be released after the Preset move is completed
	Privacy Mode	On/ <u>Off</u>	When the camera is turned off via the remote control or software, the lens will automatically turn to the bottom right to ensure the user's privacy
	Protocol	<u>VISCA</u>	Supports VISCA protocol
	Baud Rate	<u>9600</u> /38400	Transmission speed of the control signal
	VISCA Address	<u>0</u> -7	The camera ID address can be manually assigned.
	Output Mode	2160P 59.94 /50/29.97/25 1080p 59.94/50/29.97/25 720p 59.94/50/29.97/25 1080i 59.94/50	Set the output resolution (1080i is only available via HDMI. If selected, SDI will still output 1080p)
	Factory Reset	On/ <u>Off</u>	Engage factory default settings
Status			Display the current settings information

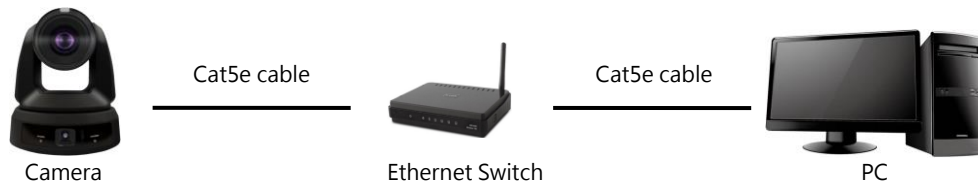
Chapter 5 Network setting and Web interface

5.1 Connecting Camera to Network

5.1.1 Connection

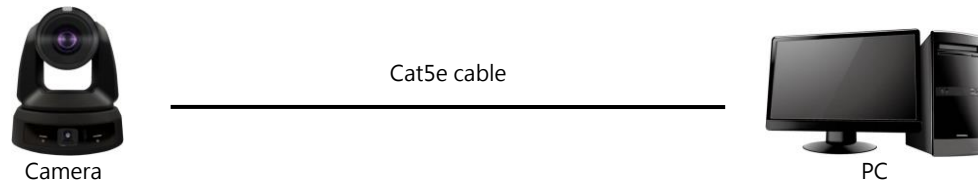
Two common connection methods are shown below

1. Connect via switch or router

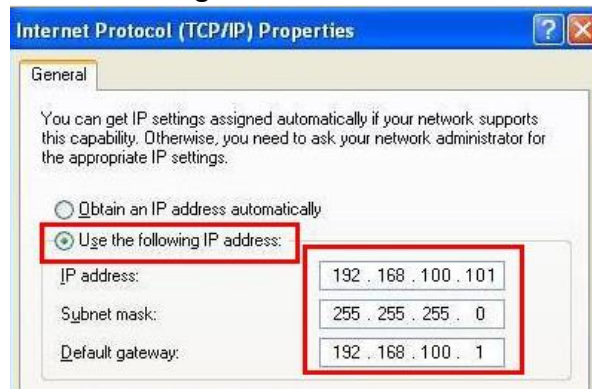


2. Connect directly via an Ethernet cable. The computer's IP address must match the network segment used by the camera.

E.g. The camera's default IP address is 192.168.100.100. The computer's IP address should be set to the same segment, such as 192.168.100.101.



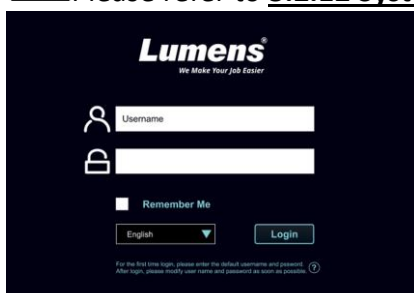
- Modify the PC's network settings



5.1.2 Use a Browser to view Images

- Open the browser, and enter the IP address of the camera in the address bar
E.g.: <http://192.168.100.100> (default IP address)
- Enter the Username and Password

Note Please refer to [5.2.12 System-User](#) to change the default password.



Lumens IP Camera

- Account: admin
- Password: 9999

5.1.3 Use an RTSP Player to View Live Images

Free software can be used for RTSP connection, such as VLC, Quick Time or PotPlayer. RTSP connection address formats are as follows:

- RTSP Stream 1 => <rtsp://camera IP:8557/h264>
- RTSP Stream 2 => <rtsp://camera IP:8556/h264>
- Panoramic Camera Output => <rtsp://camera IP:8553/h264>
<rtsp://camera IP:8553/hevc> (Can only output with hevc when the resolution of the panoramic is set to 4K)

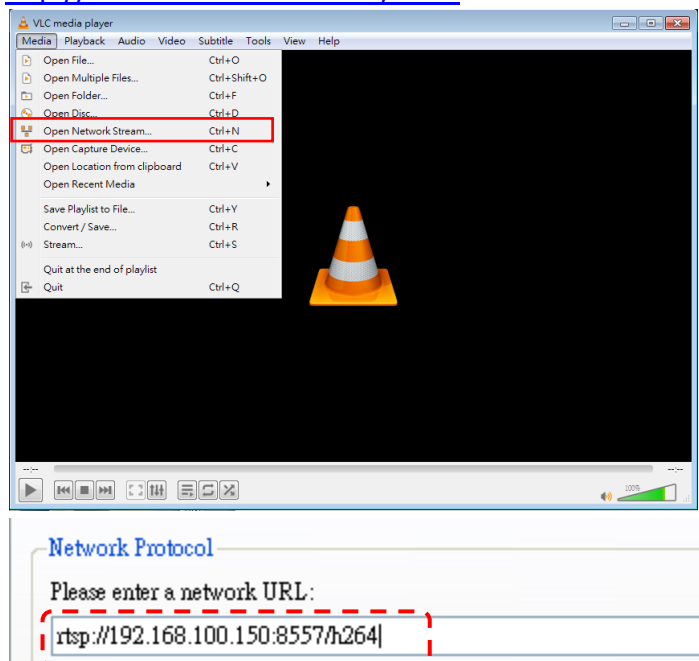
If password authentication is enabled, the RTSP connection address is as follows:

- [rtsp://Username:Password@VC IP address:8557/stream1\(or stream2\)](rtsp://Username:Password@VC IP address:8557/stream1(or stream2))
- To enable password authentication function, please refer to **5.2.10 System-Network**

Example:

Open the VLC software, click [Open Network Streaming], and enter the URL:

<rtsp://192.168.100.150:8557/h264>



5.1.4 Use Snapshot URL to View Camera Images

Snapshot URL can be used to retrieve the current camera image.

The Snapshot URL is as follows:

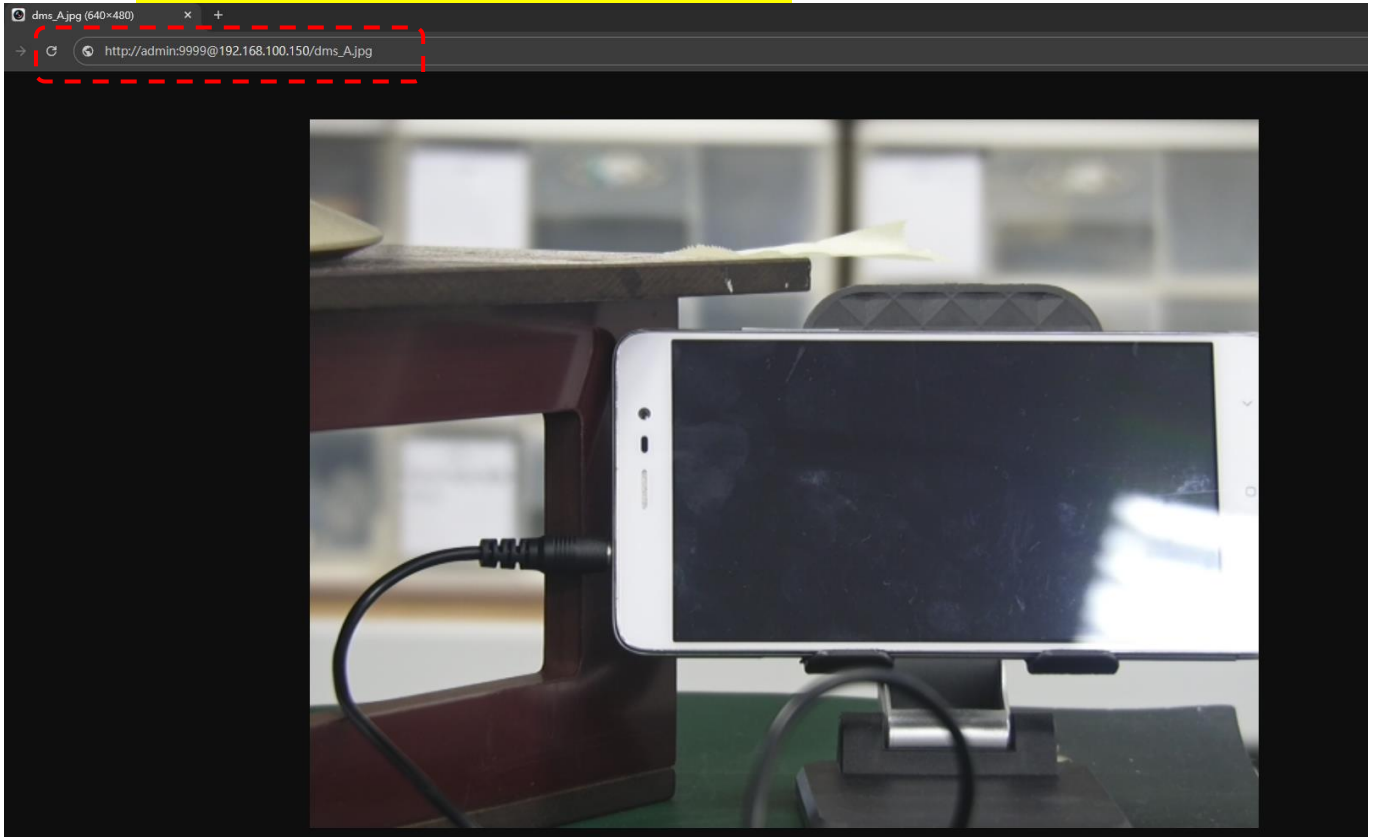
- http://user:pass@<Camera IP>/dms_A.jpg

Example:

Open a web browser and enter the following URL:

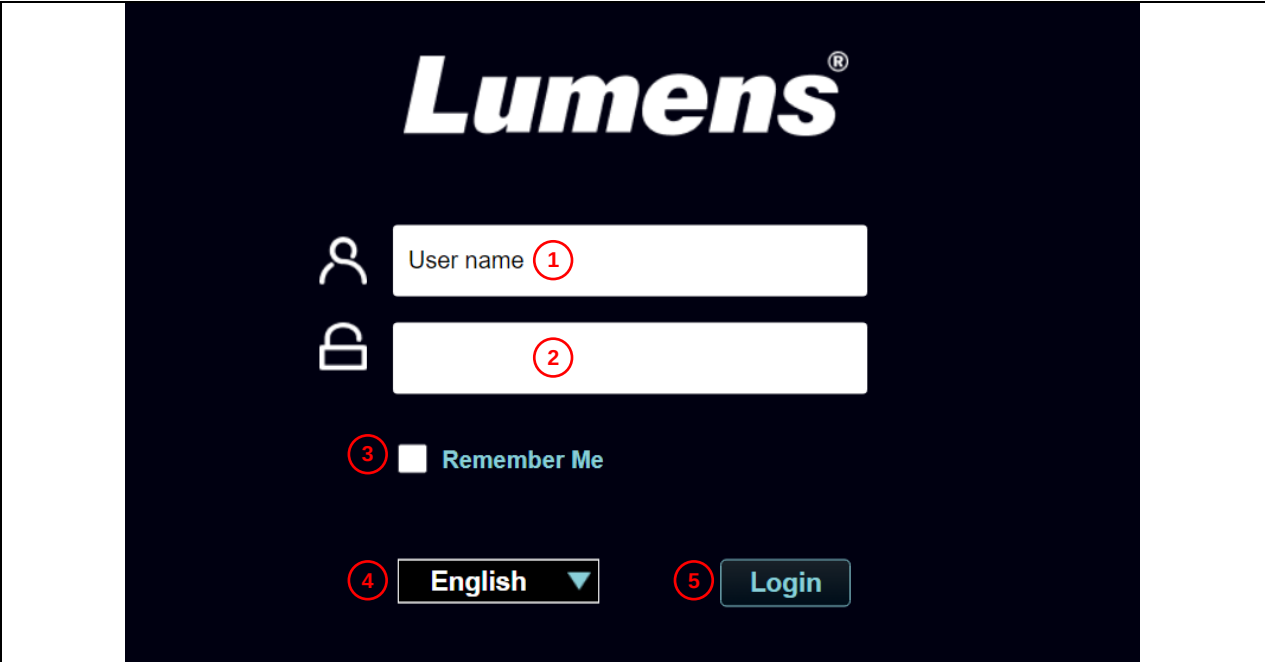
- http://admin:9999@192.168.100.150/dms_A.jpg

The current camera image will be displayed.



5.2 Web interface functions

5.2.1 Login



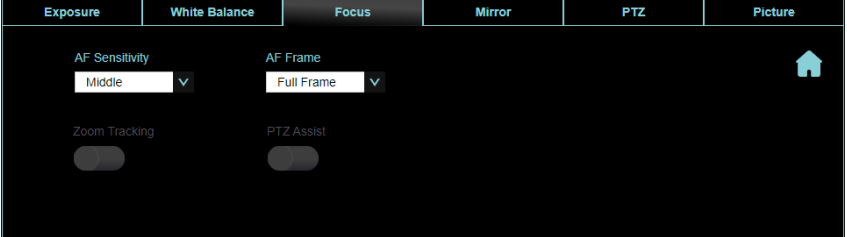
No	Item	Function Descriptions
1	User name	Enter user login credentials (default: admin)
2	Password	Enter user password (default: 9999) Note Please refer to 5.2.12 System-User to change the default password
3	Remember Me	Save the user name and password. When you log in next time, there is no need to re-enter them
4	Language	Supports English/ Traditional Chinese/ Simplified Chinese
5	Login	Log into the web interface

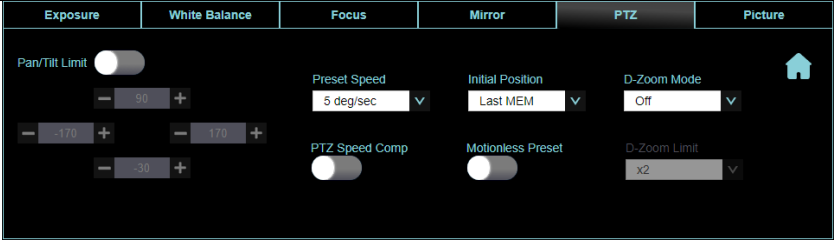
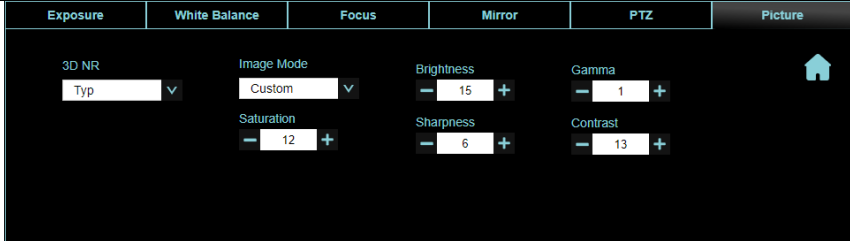
5.2.2 Live View



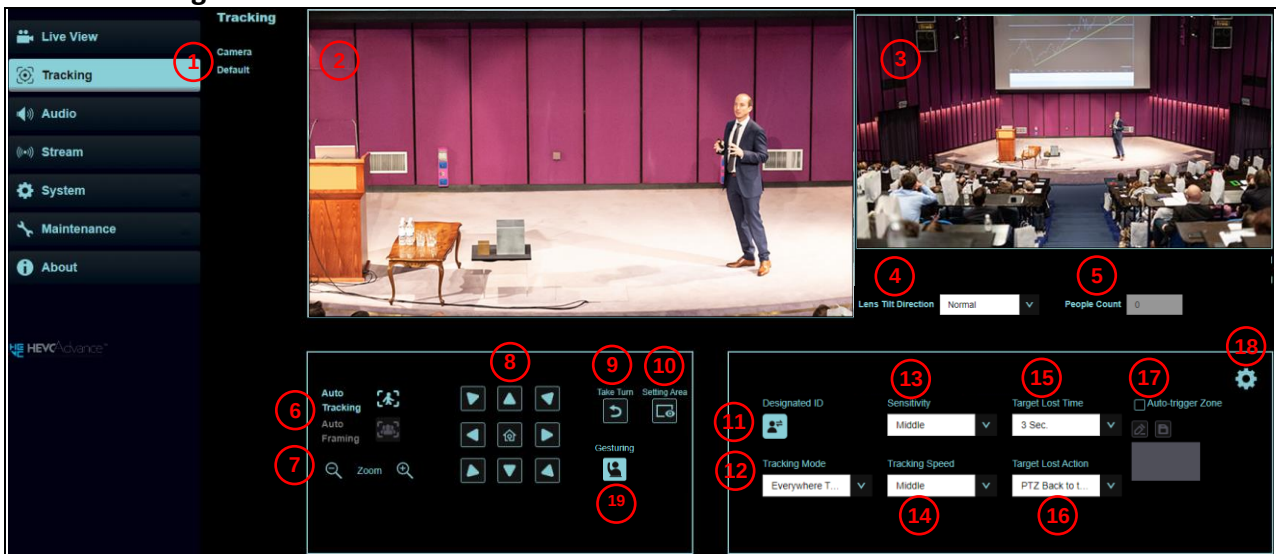
No	Item	Function Descriptions
1	Camera ID/ Location	Display camera ID/ Location Please refer to 5.2.8 System-Device
2	Preview screen	Display the live view of the camera
3	Preset setting	Select the number first and then select Store or Load
4	Pan/Tilt	Adjust the Pan/Tilt position of the camera
5	Zoom	Zoom In/Zoom Out
6	AF/ MF	Switch Auto Focus/ Manual Focus.
7	Pan/ Tilt Speed	Adjust the pan/ tilt speed
8	Zoom speed	Adjust the zoom speed
9	Tracking	Enable/ Disable Auto Tracking/ Auto Framing Note Select Auto Tracking or Auto Framing
10	Camera Setting	Please refer to 5.2.3 Live View - Camera Setting for related settings

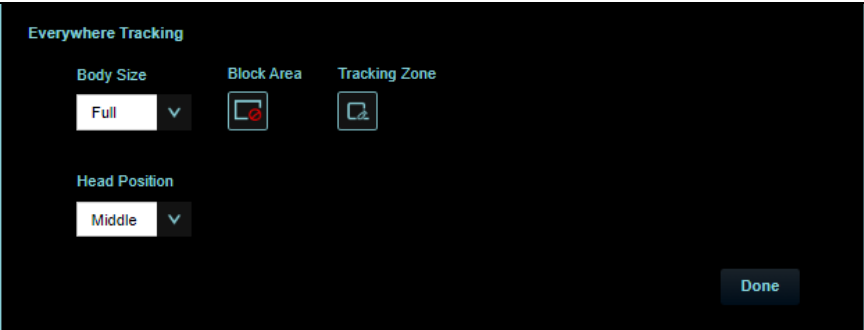
5.2.3 Live View - Camera Setting

		
No	Item	Descriptions
1.	Exposure	■ Mode: Exposure mode (Full Auto/ Shutter Pri/ Iris Pri/ Manual/ White Board) ■ WDR: set the level to change the dynamic range of the picture ■ Exposure Comp. Level: select exposure compensation level ■ Gain Level: adjust gain level (adjustable under “Manual” mode.) ■ Iris Level: adjust the aperture (adjustable under “Manual” or “Iris Priority” mode) ■ Shutter Speed: adjust the shutter speed (adjustable under “Manual” or “Shutter Priority” mode) ■ Anti-Flicker: Activate to eliminate flicker from lighting ■ HDR: Enable/Disable HDR (HDR is not supported when the resolution is set to 59.94 or 50 fps).
2.	White Balance	 ■ Mode: Select the color temperature ■ One Push: Executes a one-time color temperature adjustment (configurable in [One Push WB] mode) ■ Manual Red/Blue: Manually adjust red/blue color temperature
3.	Focus	 ■ AF Sensitivity: change the sensitivity of the auto focus. ■ AF Frame: Changes the focusing zone to the selected mode. ■ Zoom Tracking: The focus mode during lens zooming (can be set in “Manual”) ■ PTZ Assist: Available when Focus is in Manual Mode Note PTZ Assist is triggered only when the camera has moved in any axis. It will not dynamically respond to changes in the scene.
4.	Mirror	 ■ Mirror: Set image mirroring ■ Flip: Set the mode at which the image is turned

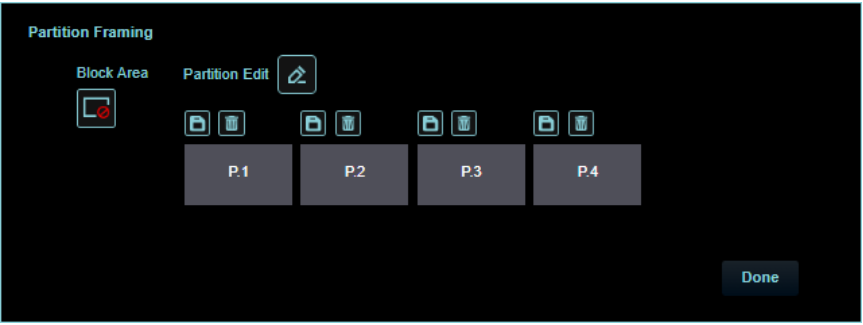
5.	Pan Tilt Zoom (PTZ)	 ■ Pan/Tilt Limit: Enable to set the pan/tilt limit ■ Pan/Tilt Flip: Set On to reverse the Pan/Tilt direction. ■ Preset Speed: Set the rotation speed of the robotic head when a Preset is executed ■ Initial Position: Set the camera lens to return to the Last MEM or the 1st Preset after POWER ON Note: 1st Preset= Preset 0 ■ D-Zoom Mode: Enable/ Disable digital zoom ■ D-Zoom Limit: Set the digital zoom limit Note: Stream/ USB/ 4K resolution does not support digital zoom ■ PTZ Speed Comp: The Pan/Tilt robotic speed will vary according to the zoom position ■ Motionless Preset: Image will freeze when a preset is activated. Freeze will be released after a Preset camera move has been completed
6.	Picture	 ■ 3D noise reduction: 3D noise reduction settings ■ Image Mode: Users can customize image parameters ■ Brightness: Adjustable when Image Mode is set to Custom ■ Gamma: Adjustable when Image Mode is set to Custom ■ Saturation: Adjustable when Image Mode is set to Custom ■ Sharpness: Adjusts the sharpness level ■ Contrast: Adjustable when Image Mode is set to Custom
7.	PTZ Control	Go back to PTZ Control page

5.2.4 Tracking

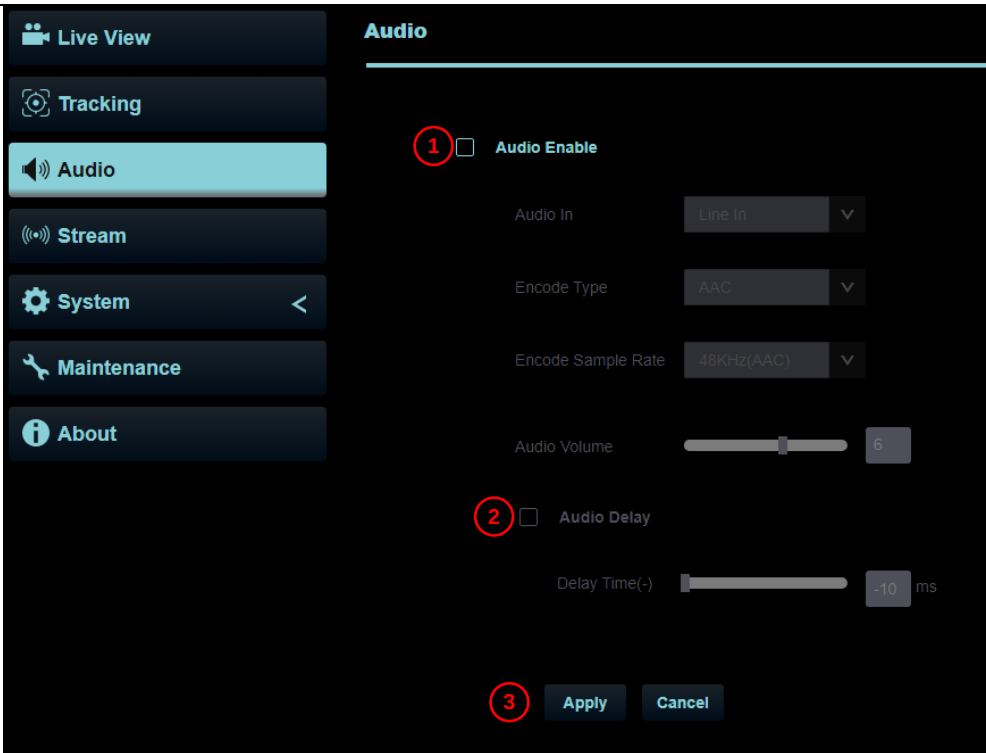
		
No	Item	Descriptions
1.	Camera ID/Location	Display camera ID/ location. Please refer to 5.2.8 System-Device

2.	PTZ preview	Display the live view of the main lens.
3.	Panoramic preview	Display the live view of the panoramic lens
4.	Lens Tilt Direction	Set the tilt position for the panoramic lens Note Ensure the setting matches the actual installed tilt angle of the camera. Please refer to 3.2.5 Tilting the Panoramic Lens
5.	People Count	Display the number of people visible to the panoramic camera
6.	Tracking	Enable/ Disable Auto Tracking/ Auto Framing Note ▪ Select Auto Tracking or Auto Framing ▪ Not applicable with [Center Stage] mode
7.	Zoom	Zoom In/Zoom Out image
8.	Pan/ Tilt	Adjust the Pan/Tilt of the camera Note PTZ setting is not supported when tracking is enabled
9.	Take turn	Switch the target Note Applicable in [Auto Tracking] mode only
10.	Setting Area	Display the configured tracking and blocked areas in the panoramic preview screen. Note Applicable in [Auto Tracking] mode only
11.	Designated ID	Display individuals the camera has detected Target ID can be switched by clicking on a subject with the mouse Note Not applicable in [Center Stage] mode
12.	Tracking Mode	Set the Tracking Mode ▪ Voice Tracking: Voice Tracking ▪ Auto Tracking: Everywhere Tracking/ Stage Tracking/ Partition Tracking ▪ Center Stage ▪ Auto Framing: Auto Framing/ Partition Framing
13.	Sensitivity	Set tracking sensitivity
14.	Tracking Speed	Set tracking speed
15.	Target Lost Time	Set a delay to take action after a tracking target is lost
16.	Target Lost Action	Set the action after a tracking target is lost
17.	Auto-trigger zone	After a person enters the area for 3 seconds, tracking/ framing will be triggered Note ▪ [Tracking Mode]> [Center Stage] is not supported. ▪ Enabling Auto Tracking/ Auto Framing will disable the [Auto-trigger zone] function ▪ Stopping Auto Tracking/ Auto Framing will reactivate the [Auto-trigger zone] function
18.	Tracking Mode Setting	Advanced settings for tracking
18.1	Everywhere tracking	 ■ Body Size: Set Full/ Half Body ■ Head Position: Set the head position of the person in the screen ■ Block Area: Click the left mouse button to draw a frame on the panoramic view to set a blocked area ■ Tracking Zone: Click the left mouse button to draw a frame on the panoramic view to set a designated tracking area

18.2	Stage Tracking	 ■ Body Size: Set Full/ Half Body ■ Head Position: Set the head position of the person in the screen ■ Block Area: Click the left mouse button to draw a frame on the panoramic view to set a block area ■ Tracking Zone: Click the left mouse button to draw a frame on the panoramic view to set a designated tracking area
18.3	Partition Tracking	Set Partition Tracking. Up to 4 zones can be set When a person enters the specific zone, the camera switches to the partition. Tracking will not be performed within the zone. Outside the zone, everywhere tracking will be active.  ■ Head Position: Set the head position of the person on the screen ■ Block Area: Click into the Edit mode, frame by holding the left mouse button and set the blocked area ■ Block Area: Click the left mouse button to draw a frame on the panoramic view to set a blocked area ■ Partition Edit: Edit and save partition positions
18.4	Auto Framing	 ■ Block Area: Click to enter the Edit mode. Frame by holding the left mouse button and set the block area

18.5	Partition Framing	 ■ Block Area: Click the left mouse button to draw a frame on the panoramic view to set a block area ■ Partition Edit: Edit and save partition positions
19	Gesturing	When Gesture Control is enabled, the camera will automatically move to the person who raises their hand during tracking. Limitations: 1. Only a palm or fist gesture is supported. 2. The hand must be raised above shoulder level. 3. The distance from the camera must not exceed 8 meters.

5.2.5 Audio

		
No	Item	Descriptions
1	Audio Enable	Select to enable Audio function ■ Audio In: Line In ■ Encode type: AAC ■ Encode sample rate: 48KHz (AAC) ■ Volume: Adjust Audio Volume
2	Audio Delay	Check to set the audio delay time (10 - -500 ms)
3	Apply/ Cancel	Apply/ Cancel Audio Settings

5.2.6 Stream

Note NDI function is only applicable to the VC-TR61N.

No	Item	Descriptions
1	Stream	Supports 2 streams/ NDI/ Panoramic live stream
2	Stream settings	Please refer to 5.2.6.1 Stream Settings .
3	RTSP	Select to enable RTSP ■ Enable/ Disable Multicast Multicast is recommended when the stream is watched by more than 4 viewers ■ Enable/Disable Password Authentication ➢ Please refer to 5.1.3 Using RTSP Player to View the Images for the connection format ➢ The username/ password is the same as the web login password of the camera. Refer to 5.2.12 System Settings - User to add/modify account information
4	RTMP/ RTMPS	Use the RTMP URL provided by the RTMP service platform to publish the camera live view For YouTube streaming, refer to 5.2.5 Audio to turn on the audio function
5	SRT	■ Check to enable and set SRT streaming After the SRT streaming is opened, it will be connected automatically upon camera startup ■ The port number must be set in the range above 1024 with a maximum value of 65535 Note The following ports are used by the camera. Setting the ports may result in connection issues. 8554, 8556, 8557, 8080, 9090, 1935 ■ Latency can be set between 20-8000 microseconds. (Default is 120)

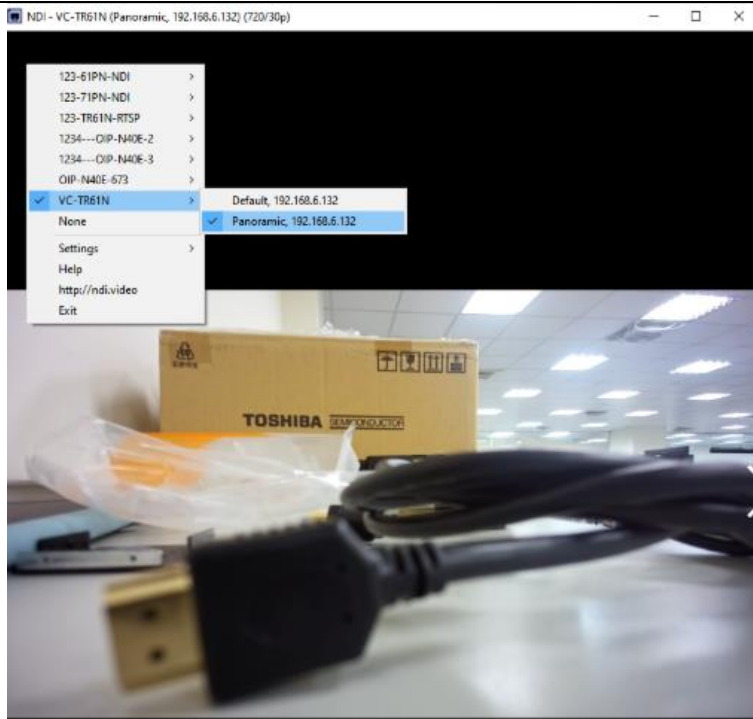
		microseconds
--	--	--------------

5.2.6.1 Streaming settings

Function		Stream 1	Stream 2
Encode Format		H.264	H.264
Resolution		4K/ 1080p/ 720p	720p
Bit rate	Bit rate	2,000-20,000	2,000-20,000
	Default	7,000	3,000
Rate Control		CBR / VBR	
Frame Rate/ Group of Pictures		Setting according to the supported resolution	

5.2.7 Stream – NDI (For VC-TR61N only)

No	Item	Function Descriptions
1	Camera ID/ Location	Displaying ID/location according to [System] > [Output] settings
2	Group Name	The group name can be modified here and set with Access Manager - Receive in NDI Tool ■ If no group is assigned, the machine belongs to Public. As the default of Access Manager includes Public, Studio Monitor can discover this machine. ■ If a group is assigned, but this group is not added into Access Manager, Studio Monitor cannot discover this machine. ■ If a group is assigned and this group is added into Access Manager, Studio Monitor can discover this machine. Note The name can contain commas (,) to distinguish different groups Example: "default, 123, abc" means this machine belongs to three groups at the same time (default/ 123/ abc)
3	NDI HX	HX2/ HX3 is supported After selecting NDI HX2 or HX3, the VC-TR61N can output both the Panoramic and PTZ scenes simultaneously over NDI. In NDI Studio, you can select either one.

		
4	Multicast	Multicast Settings. It is recommended to enable this feature when there are more than 4 viewers online simultaneous
5	Discovery Server	Check to enter the Server IP address
6	Apply	After the setting has been modified, click this button to apply the settings

5.2.8 System - Device

Live View

Tracking

Audio

Stream

System ▼

Device

Output

Device

Camera ID

VC-TR61

Location

Default

Apply

Cancel

Descriptions

Edit camera ID/ Location

- Camera ID: Supports 1 - 32 characters
- Location: Supports 1 - 30 characters

- Supports English, Chinese and numbers. Special symbols such as “/” and “space” cannot be used
- Note** Modifying this field will also update the ONVIF device name/location.

5.2.9 System - Output

No	Item	Descriptions
1	Resolution	Set the resolution of the camera (1080i is only available via HDMI. If selected, SDI will still output 1080p) Note After changing resolution, the camera will restart. The browser will need to be refreshed
2	HDMI Output Source	Configure PTZ/ Panoramic Output
3	HDMI Format	Supporting YUV422/ RGB
4	Privacy Mode (UVC)	After the privacy mode is enabled, when the camera is turned off via the remote control or software, the lens will automatically turn to the bottom right to ensure the user's privacy
5	Panoramic 4K Mode	Supports 4K output Note ■ This mode is available when the resolution is set to 4K 60/30fps ■ UVC is limited to 720p when Panoramic 4K mode is selected ■ After changing resolution, the camera will restart. The browser will need to be refreshed
6	PIP	Enable/Disable PIP(Picture-in-Picture) ■ Mode Select: PTZ / Panoramic Sub Screen ■ Size: Set the size of the sub-screen ■ Position: Set the display position of the sub-screen
7	Apply/ Cancel	Apply/ Cancel Setting

5.2.10 System - Network

Live View

Tracking

Audio

Stream

System

Output

Network

Date & Time

User

Control

Maintenance

About

Network

1

DHCP

IPv4 Address

192.168.4.115

IPv4 Netmask

255.255.255.0

Gateway

192.168.4.254

DNS

192.168.1.211

MAC Address:

dc:e2:ac:03:e0:50

2

HTTP Port

80

3

HTTPS Port

81

4

Apply

Cancel

No	Item	Descriptions
1	DHCP	Camera network settings. Configurable when DHCP is set to Off
2	HTTP Port	Set HTTP port. The default Port value is 80
3	HTTPS Port	Set HTTPS port. The default Port value is 81
4	Apply/ Cancel	Apply/ Cancel Setting

5.2.11 System - Date & Time

Live View

Tracking

Audio

Stream

System

Output

Network

Date & Time

User

Date & Time

Current Date & Time

2024-07-26 10:53:34

PC Clock

2024-07-26 10:53:36

Date & Time Format

yyyy-mm-dd hh:mm:ss

Time Setting

Synchronize with PC

Apply

Cancel

Description
Display the current camera/PC date/time, and set the display format. Manually set [Time Settings], Date & Time or synchronize with your PC.

5.2.12 System - User

Lumens®

29

Live View

Tracking

Audio

Stream

System ▾

Device

Output

Network

Date & Time

User

Control

User

EDIT

DEL

User Name	Authorization Mode
☐ admin	Administrator
☐ +	

Description

Add/Modify/Delete user account

- User name and password supports 4 - 32 characters
- Use uppercase, lowercase letters or numbers. Special symbols cannot be used
- Account permissions:

User Type	Admin	Operator	Viewer
Live View	V	V	V
Settings	V	V	X
Account management	V	X	X

Note

Performing a factory reset will clear all user data.

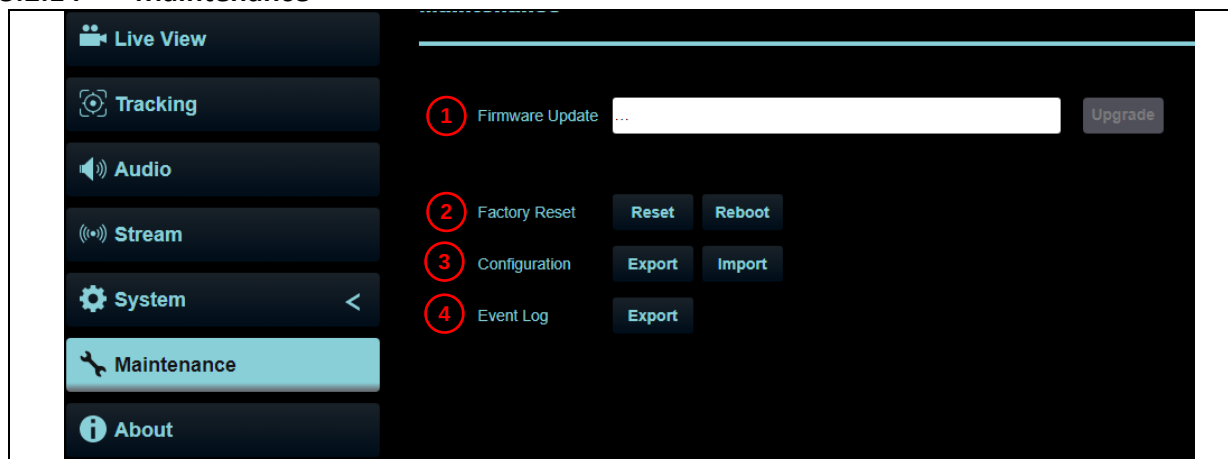
5.2.13 System - Control

The screenshot displays the 'Control' interface of a Lumens device. On the left is a sidebar with menu items: Live View, Tracking, Audio, Stream, System (selected), Output, Network, Date & Time, User, and Control. The main area is titled 'Control' and contains several settings, each numbered in a red circle:

- 1 Protocol: VISCA
- 2 Baud Rate: 9600
- 3 VISCAAddress: 1
- 4 Tracking Led Status: Off
- 5 Tally Lamp: Disable
- 6 Tally CMMD Mode: Link
- 7 Apply/Cancel buttons

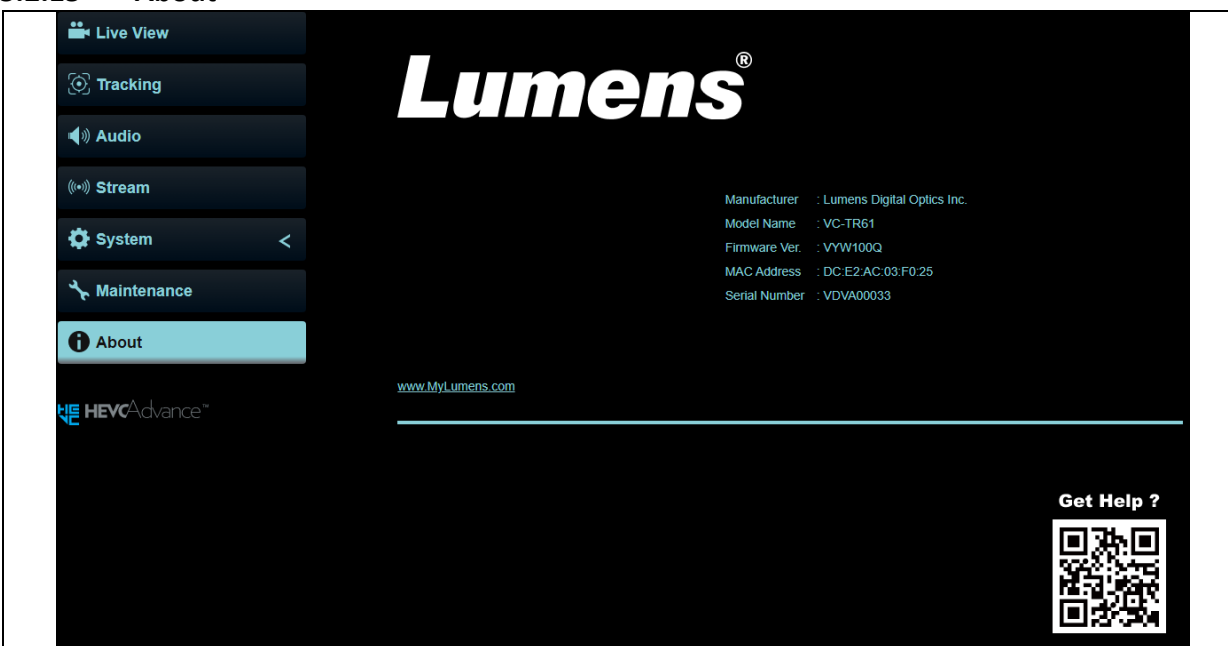
No	Item	Function Descriptions
1	Protocol	Supports VISCA protocol
2	Baud Rate	Select the transmission speed of the control signals
3	VISCA address	The camera ID address 0 - 7 can be assigned
4	Tracking LED status	When enabled, the LED on the front panel will flash green during tracking
5	Tally Lamp	Enable/ disable the tally lamp
6	Tally CMMD Mode	Set whether the tally lamp is linked with RS232 commands ■ Normal (Not linked): The tally lamp and mode are set separately ➤ When the tally lamp is on but the default mode is off, an additional command is required to set the indicator light mode ➤ When the tally lamp is off, setting the mode will not trigger the indicator light ■ Link: ➤ When the tally lamp is on, the default mode is red light ➤ When the tally lamp is off, setting the mode will turn on the tally lamp.
7	Apply/ Cancel	Apply/ Cancel Setting

5.2.14 Maintenance



No	Item	Descriptions
1	Firmware Update	Click [...] to select the firmware file and click [Upgrade] to start updating Note - The update process takes 2-3 minutes - Do not operate or turn off the power of the device during the update to avoid firmware failure
2	Factory Reset	- Reset all configurations to default settings - Reboot the camera
3	Configuration	Save parameters as a Profile. Users can share camera setup parameters
4	Event Log	If the camera operates abnormally, please export event log and provide it to Lumens for investigation

5.2.15 About



Description
Display the firmware version, serial number, and other related information. For technical support, please scan the QR code at the bottom right.

Chapter 6 Troubleshooting

This chapter describes problems you may encounter while using VC camera. If you have questions, please refer to related chapters and follow all the suggested solutions.

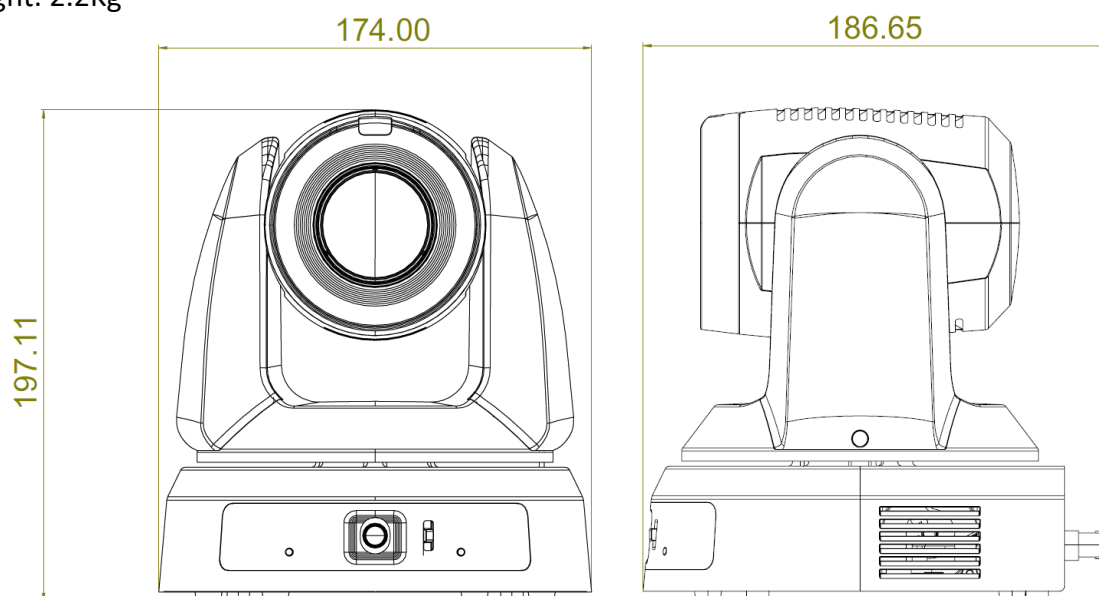
If the problem still occurs, please contact your local distributor or Lumens.

No.	Problems	Solutions
1.	No power to the device	1. Make sure you have plugged in the power cord 2. When using a PoE connection, ensure that the power supply supports PoE+ (IEEE802.3at)
2.	There is no image output from camera	1. Check the power supply or PoE supply functions 2. Confirm whether the camera resolution can be used with the connected display ▪ If the camera is set to 4K resolution, output, confirm the display supports 4K output 3. Replace the cable and make sure it's not faulty
3.	Camera image is severely delayed	Use 60/50 fps rather than 25/30 fps signals
4.	RS-232 cannot be controlled	1. Confirm the connection is correct (RS-232 In/ Out) 2. Please make sure the Baud rate setting is the same as the control device
5.	ONVIF software cannot discover the camera	Please make sure that [System]> [Device]> [Camera ID]/ [Location] in the web page uses only English letters or numbers. The ONVIF software cannot find the camera if you use special symbols or space
6.	The camera does not save relevant parameters (PTZ, AWB ...) after reboot	Please make sure if the [Initial Position] is set to [Last MEM] from the setting menu or web page ▪ OSD Menu: [System]> [Initial Position] ▪ Web page: [Live View]> [Setting] > [PTZ]> [Initial Position]
7.	How to log into the webpage if you have forgotten the password	Please contact your local distributor or Lumens technical support
8.	Admin password when deleting the User Account	Please contact your local distributor or Lumens technical support
9.	No image on SDI output when resolution is set to 1080i	The TR61 does not support 1080i output over SDI. If 1080i is selected, the SDI port will continue to output 1080p. If 1080i is required, consider using an HDMI-to-SDI converter to output the image from the HDMI port.

Chapter 7 Addendum

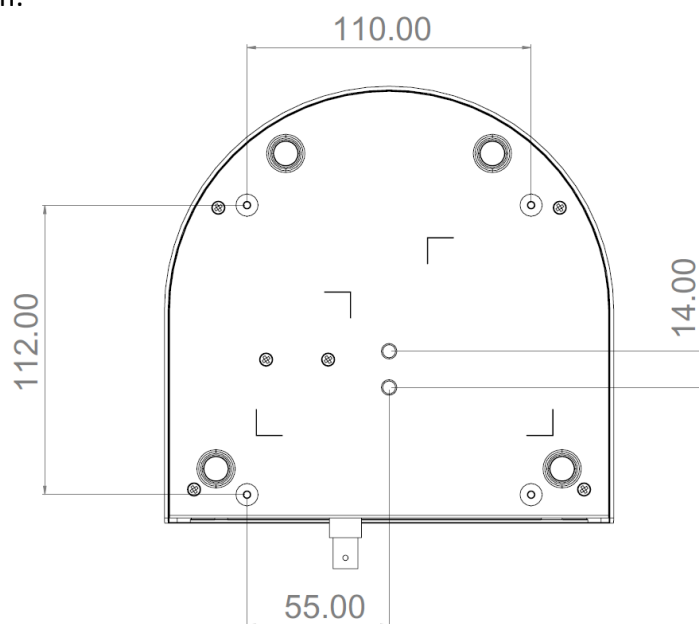
7.2 Camera Size

- Length x Width x Height: 174 x 186.65x 197.11mm
- Weight: 2.2Kg



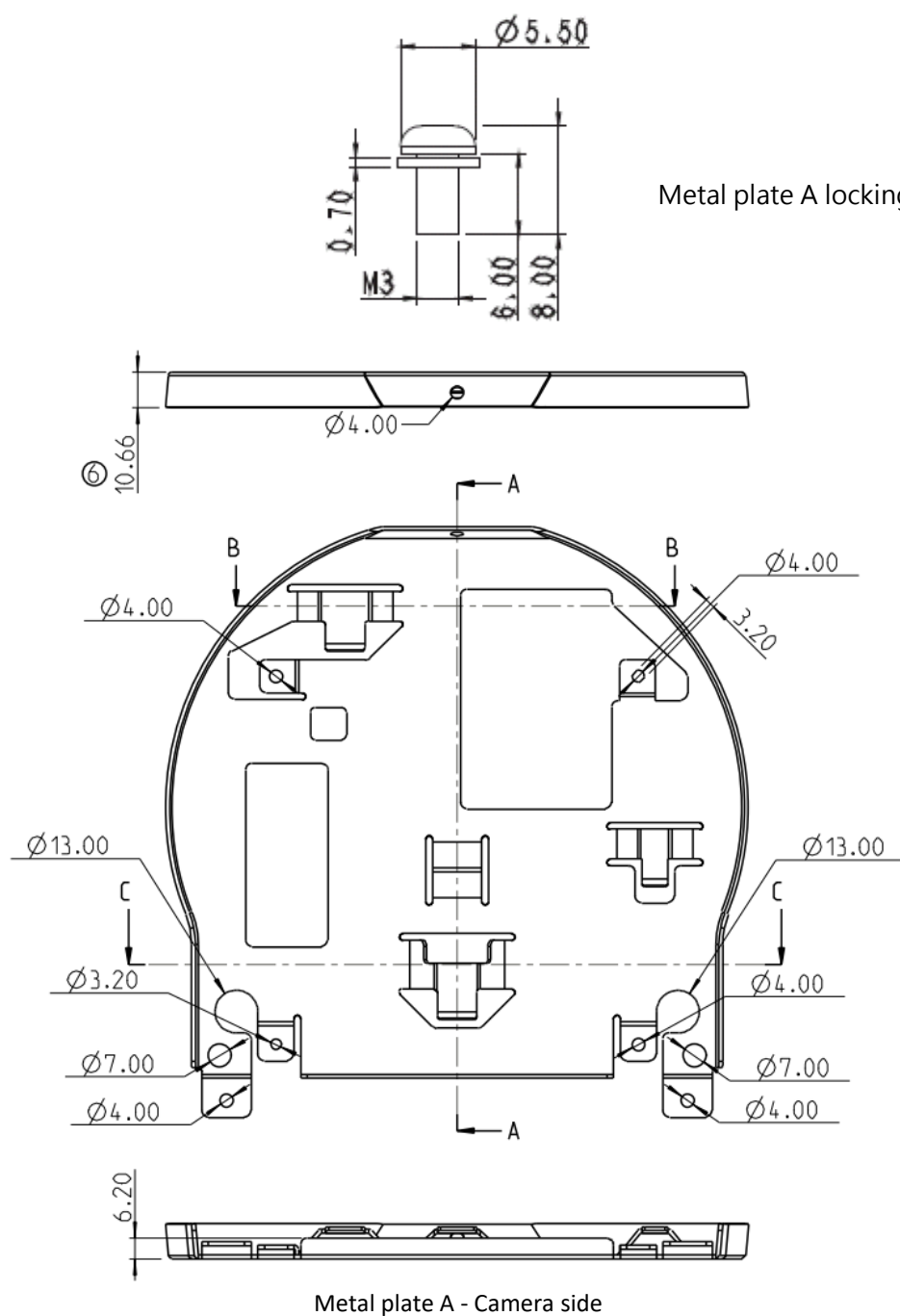
7.3 Dimensions of the base

The camera can be mounted on a 1/4"-20 UNC PTZ tripod head by using the locking holes on the bottom.



7.4 Metal Plate diagram

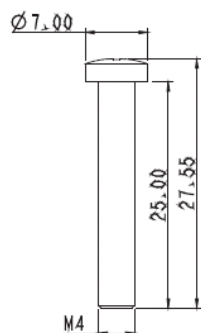
(1) Metal plate A - camera side



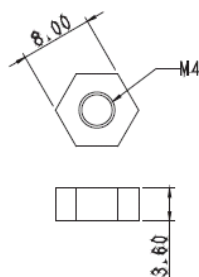
Metal plate A locking screw

Metal plate A - Camera side

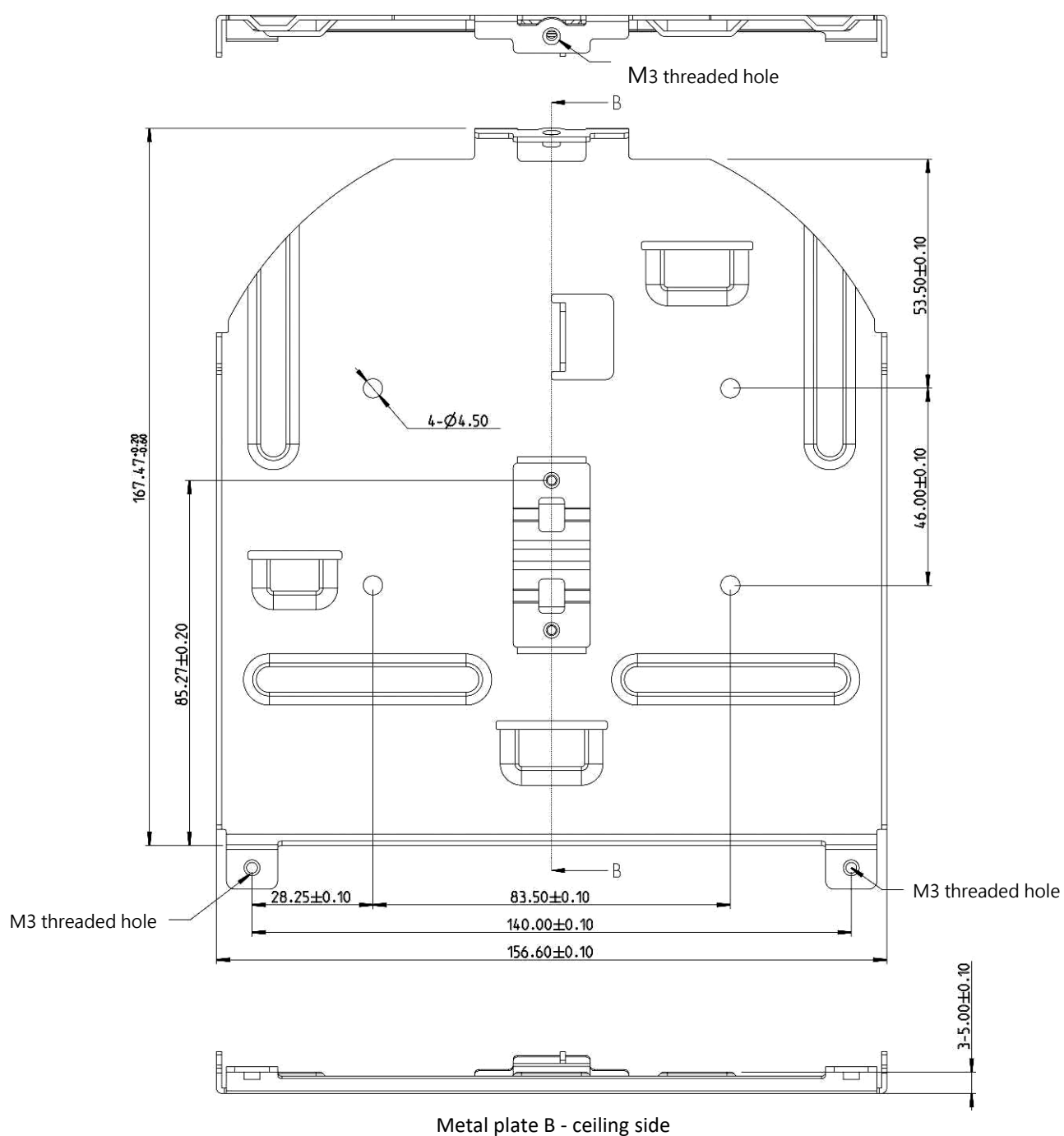
(2) Metal plate B - ceiling side



Metal plate B locking screw



Metal plate B locking bolt



Metal plate B - ceiling side

Chapter 8 Safety Instructions

Always follow these safety instructions when setting up and using PTZ Video Camera:

1 Operation

- 1.1 Please use the product in the recommended operating environment, away from water or source of heat.
- 1.2 Do not place the product on a tilted position or unstable trolley, stand or table.
- 1.3 Please clean the dust on the power plug prior to usage. Do not insert the product's power plug into a multiplug to prevent sparks or a fire.
- 1.4 Do not block the slots and openings in the case of the product. They provide ventilation and prevent the product from overheating.
- 1.5 Do not open or remove covers, otherwise it may expose you to dangerous voltages and other hazards. Refer all servicing to licensed service personnel.
- 1.6 Unplug the product from the wall outlet and refer servicing to licensed service personnel when the following situations happen:
 - If the power cords are damaged or frayed.
 - If liquid is spilled into the product or the product has been exposed to rain or water.

2 Installation

- 2.1 For security considerations, please make sure the standard hanging rack you bought is in line with UL or CE safety approbations and installed by technician personnel approved by agents.

3 Storage

- 3.1 Do not place the product where the cord can be stepped on as this may result in fraying or damage to the lead or the plug.
- 3.2 Unplug the product during thunderstorms or if it is not going to be used for an extended period.
- 3.3 Do not place the product or accessories on top of vibrating equipment or heated objects.

4 Cleaning

- 4.1 Disconnect all the cables prior to cleaning and wipe the surface with a dry cloth. Do not use alcohol or volatile solvents for cleaning.

■ Precautions

	This symbol indicates that this equipment may contain dangerous voltage which could cause electric shock. Do not remove the cover (or back). No user-serviceable parts inside. Refer servicing to licensed service personnel.		WARNING
			This symbol indicates that there are important operating and maintenance instructions in this User Manual with this unit.

■ Battery Warning (for products or accessories with batteries)

General Safety Warnings:

- Do not heat or dispose of the battery in fire, as it may explode or leak.
- Avoid short-circuiting: Do not connect the positive and negative terminals directly or use metallic objects to connect them.
- Do not disassemble, crush, puncture, or modify the battery.
- Do not immerse the battery in water or any other liquids.

Usage and Storage Warnings:

- Do not store the battery in high-temperature environments (above 60°C) or in areas with high humidity.
- Keep away from direct sunlight and heat sources.
- If not in use for a long period, remove the battery and store it in a cool, dry place.

Disposal and Recycling Warnings:

- Do not dispose of the battery in household waste; follow local regulations for proper disposal and recycling.
- If leakage occurs, avoid contact with the leaked substance and clean the affected area immediately.

Child Safety Warnings:

- This product contains a button battery. Ingestion of a button battery can cause severe internal burns within 2 hours and may result in death.
- If a battery is swallowed or if internal battery material is contacted, seek medical attention immediately.
- Store batteries (whether new or used) in a place that is out of reach of children.
- If the battery compartment does not close securely, stop using the product and keep it away from children.

■ FCC Warning

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

Notice :

The changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

■ IC Warning

This digital apparatus does not exceed the Class A limits for radio noise emissions from digital apparatus as set out in the interference-causing equipment standard entitled "Digital Apparatus," ICES-003 of Industry Canada.

Cet appareil numérique respecte les limites de bruits radioélectriques applicables aux appareils numériques de Classe A prescrites dans la norme sur le matériel brouilleur: "Appareils Numériques," NMB-003 édictée par l'Industrie.

■ EN55032 CE Warning

Operation of this equipment in a residential environment could cause radio interference.

Copyright Information

Copyrights © Lumens Digital Optics Inc. All rights reserved.

Lumens is a trademark registered by Lumens Digital Optics Inc.

Copying, reproducing or transmitting this file is not allowed if a license is not provided by Lumens Digital Optics Inc. unless copying this file is for the purpose of backup after purchasing this product.

In order to keep improving the product, the information in this file is subject to change without prior notice.

To fully explain or describe how this product should be used, this manual may refer to names of other products or companies without any intention of infringement.

Disclaimer of warranties: Lumens Digital Optics Inc. is neither responsible for any possible technological, editorial errors or omissions, nor responsible for any incidental or related damages arising from providing this file, using, or operating this product.